

CLAIRE MERCIER

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PROFESSIONAL SUMMARY

I'm a Research Engineer with 7 years of experience developing machine learning systems and distributed computing solutions. I've spent the last four years at innovative tech companies in Paris, working on large-scale data processing and AI research projects. I'm particularly drawn to solving complex engineering problems that bridge research prototypes and production systems. Outside work, I enjoy contributing to open source projects and mentoring junior engineers.

WORK EXPERIENCE

Senior Research Engineer

Neuron Labs

www.neuronlabs.io

Paris, France

March 2022 - Present

Led the design and implementation of a distributed training framework for large language models serving 15+ concurrent research projects across the organization. Optimized GPU memory utilization, reducing training costs by 40% through custom kernel development in CUDA. Established best practices for reproducible research by building containerized experiment pipelines with automated versioning and artifact management. Mentored a team of 3 junior engineers on distributed systems architecture and deep learning optimization techniques.

Research Engineer II

DataFlow Systems

www.dataflow-systems.net

Paris, France

September 2019 - February 2022

Developed and maintained an Apache Spark-based ETL pipeline processing 2TB of financial data daily for risk analysis and reporting. Implemented a custom Python library for feature engineering that reduced model training time by 35% and became standard across the data science team. Designed and conducted performance benchmarks comparing TensorFlow and PyTorch for a critical recommendation system, influencing technology stack decisions. Collaborated with research scientists to productionize novel anomaly detection algorithms, handling edge cases and scaling challenges.

Research Engineer

Institut Technologie Avancée

www.ita-paris.co

Paris, France

July 2017 - August 2019

Built end-to-end computer vision pipelines for real-time object detection using YOLOv3, deployed on edge devices with inference times under 50ms. Created comprehensive documentation and conducted workshops to help researchers transition from TensorFlow 1.x to 2.x across the organization. Developed automated testing frameworks for model validation that caught production bugs in 3 critical experiments before deployment. Participated in research publication process, contributing to 2 peer-reviewed papers on optimization techniques for

neural network inference.

EDUCATION

Master of Science in Computer Science

Université Pierre et Marie Curie (Sorbonne Université)

Paris, France

2015 - 2017

Specialization: Machine Learning and Distributed Systems

Bachelor of Science in Computer Science

Université Pierre et Marie Curie (Sorbonne Université)

Paris, France

2012 - 2015

TECHNICAL SKILLS

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, computer vision, NLP fundamentals, model optimization

Distributed Systems: Apache Spark, Kubernetes, Docker, distributed training frameworks, Ray

Programming Languages: Python, C++, SQL, Java

Data Processing: Pandas, NumPy, Polars, data pipelines, ETL design

Cloud Platforms: AWS (EC2, S3, SageMaker), Google Cloud Platform

DevOps: Git, continuous integration, containerization, monitoring and logging

Databases: PostgreSQL, MongoDB, Redis

LANGUAGES

French: Native

English: Fluent

German: Intermediate